Casey Mathieson

Electro-Mechanical Systems · Automation · Precision Manufacturing

Oregon City, OR · 503-476-5112 · caseymathieson@gmail.com · caseymathieson.github.io · linkedin.com/in/caseymathieson

Professional Summary

Electro-mechanical systems engineer with 11+ years of experience designing, building, commissioning, and improving automation equipment, manufacturing processes, and precision-metrology systems. Brings hands-on ownership from root-cause analysis and concept development through mechanical/electrical design, fabrication, testing, deployment, and durable process documentation.

Technical Expertise

Systems & metrology: precision metrology, laser calibration, sensors/instrumentation, motion and robotic systems, pneumatics, thermal control, test-method development, root-cause analysis. **Mechanical & manufacturing:** SolidWorks (CSWP), PDM, GD&T, FEA/simulation, fixtures/nests, DFM/DFA, CNC/CAM/G-code, fabrication. **Electrical & safety:** AC/DC panels, schematic review/redlines, motor control/VFDs, PLCs, machine-safety design, programming, and validation. **Process & delivery:** commissioning, FAT/SAT, calibration, PFMEA, SOPs, estimates, technical coordination, and technician-ready documentation. **Engineering tooling:** agent-assisted development and validation of Python, MATLAB, and TypeScript tools; research, workflow automation, and structured data.

Professional Experience

ZeroTouch Metrology / DWFritz Automation — Wilsonville, OR
2020–Present

Systems Engineer, ZeroTouch Segment · 2024–Present

- Drive process improvement across precision-machine buildup, commissioning, deployment, and support.
- Conduct R&D, calibration, and test-method development for precision-metrology hardware.
- Commission systems through on-site installation and testing, resolving mechanical, electrical, and software issues.
- Develop agent-assisted engineering workflow tools for research, SOP development, troubleshooting, and repeatable support tasks, with human review of technical outputs.

Mechanical Engineer · 2020–2024

- Modernized legacy machine designs and developed mechanical solutions for conveyance, packaging, part handling, and robotic end-of-arm tooling.
- Designed precision fixtures, nests, and adjustment stages; improved calibration methods, DFM, CAD/BOM quality, and technician-ready documentation.
- Developed sensor- and dewpoint-based thermal-control logic to prevent condensation damage and enable safe operating responses.

Evergreen Advancements / Cascadia Digital — Oregon
2023–Present

Founder / Operator

- Build and operate a digital-product business; apply automation, process design, and AI-assisted research tooling to improve repeatability in product-development and storefront workflows.

Sig Sauer Electro-Optics — Wilsonville, OR
Oct 2019–Feb 2020

Manufacturing Engineer

- Designed and integrated automated equipment and tooling for optical-electronics assembly; improved reliability, throughput, and safety through troubleshooting, six-axis robot programming, and PLC changes.

Production Robotics Inc. — San Leandro, CA
May 2015–Jul 2019

Senior Automation Engineer · Nov 2016–Jul 2019

Mechanical Design Engineer · May 2015–Nov 2016

- Designed and delivered custom manufacturing and assembly automation programs ranging from $30K to $1.5M.
- Performed design reviews, PFMEAs, FAT/SAT, and high-risk prototype/process testing across simultaneous projects.
- Designed mechanical assemblies and ground-up AC/DC electrical panels, including motor control, VFDs, and NFPA/NEC/NEMA-aligned schematics.
- Developed estimates, proposals, schedules, and technical delivery plans for robotic assembly and material-handling systems.

West Coast Surgical LLC — Half Moon Bay, CA
Oct 2014–May 2015

CNC Machinist

- Produced high-mix, low-volume medical-device work; optimized milling setups, G-code, and workholding fixtures to meet tolerance requirements.

Shaffer's Offroad — Alameda, CA
Jun 2014–Oct 2014

CAD Designer / Fabricator

- Designed and manufactured custom vehicle components using SolidWorks, CNC plasma, press-brake, milling, and MIG-welding processes.

NASA Ames Research Center — Moffett Field, CA
Jun 2013–Oct 2014

Engineering Intern, Volunteer

- Contributed to R&D for a cable-actuated tensegrity rolling robot and led a senior-project team that prototyped a modular motorized cable-actuation system.

Education & Credentials

B.S., Mechanical Engineering, Mechatronics Focus — San Francisco State University, 2015

A.A., Economics; A.A., Marketing, Graphic Design Focus — Santa Barbara City College, 2011

Certified SOLIDWORKS Professional — Mechanical Design — Certificate C-QNHPJK9BLV, 2019